

DEVELOPMENT OF A FINAL YEAR STUDENTS PROJECT REPOSITORY SYSTEM (A CASE STUDY OF FEDERAL POLYTECHNIC BIDA)

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ABSTRACT

The Federal Polytechnic Bida annually graduates students from its National Diploma (ND) and Higher National Diploma (HND) programs, requiring them to complete final year project. Currently, these projects are submitted as hard copies to the school library as well as the department libraries as references for future research. Often, the process of applying to review a copy of these research works is cumbersome, as students have to go through time-consuming procedures to access these documents. This paper proposes the design and implementation of an online repository system to digitize and centralize the storage of these final year projects. The proposed system will enable students to upload, view, and retrieve project works from all departments under the institution's various schools, transforming these papers into readily accessible research documents. A web-based application using PHP within the AdminLTE3 framework and MySQL for database management. This approach ensures an easy to navigate and user-friendly platform that enhances the availability of academic resources, promotes knowledge sharing, and supports academic research. The developed system not only addresses the inefficiencies and limitations of the current system but also provides a scalable, secure, and user-friendly platform for students and researchers.

Keywords: Project, Graduate, Web-Based, Repository

INTRODUCTION

The Federal Polytechnic Bida annually graduates students from its National Diploma (ND) and Higher National Diploma (HND) programs, requiring them to complete final year project papers. Currently, these projects are submitted as hard copies to the school library as well as the department library as references for future research. Often, the process of applying to review a copy of these research works is cumbersome, as students have to go through time-consuming procedures to access these documents. This paper proposes the design and implementation of an

online academic repository system to digitize and centralize the storage of these final year projects.

A final year student project is a formal document written by students to present original research, findings, reviews, or theoretical insights (Khan *et al.*, 2023). These papers undergo rigorous peer review to ensure the credibility and accuracy of the information presented. Final year student projects are essential for the advancement of knowledge in various fields, contributing to the ongoing discourse within academic and scientific communities (González *et al.*, 2020).

A repository refers to a digital archive that collects, preserves, and disseminates academic works such as journal articles, conference papers, theses, and dissertations (Daiyabu *et al.*, 2023). The purpose of an academic repository is to provide a systematic and organized way for researchers to access and share scholarly materials (Abdelrahman, 2021). This type of repository plays a crucial role in ensuring that valuable research is preserved and made accessible to the academic community and beyond. It helps in managing the large volume of research outputs generated by universities, research institutions, and individual scholars.

An online final year student project repository takes this concept further by leveraging the internet to provide global access to final year student projects (Nwachi & Idoko, 2021). Unlike traditional repositories that might require physical access or localized networks, online repositories make it possible for anyone with internet access to search, retrieve, and read final year student projects from anywhere in the world. This greatly enhances the dissemination of knowledge, fostering a more inclusive and interconnected global academic community.

OBJECTIVES OF THE RESEARCH

- i. Developing a robust database system to securely store digitized final year projects and, ensure efficient retrieval.
- ii. Designing an intuitive user interface for easy navigation and access to final year project papers.
- iii. Implementing the designed authentication mechanisms to regulate access and maintain the confidentiality of academic materials.

REVIEW OF RELATED WORKS

Adegbile *et al.*, (2018) underscores the utilization of a Project Repository System in educational administration as a viable alternative to the manual storage method for past project documents and class materials. The Federal College of Animal Health and Production Technology serves as a case study for this implementation. The focus of the Project Repository System application is on archiving past project works within the institution, aiming to mitigate the challenges such as stress, errors, loss, and damage inherent in manual record-keeping practices. Specifically tailored to address the needs for designing a project repository system, this project is presented succinctly, covering essential background information. Utilizing Visual Basic programming language and Windows form application, the software development process aims to alleviate the

difficulties associated with the previous manual system. The envisioned software solution is expected to streamline operations, rendering them more efficient, timely, and reliable. While Visual Basic programming language and Windows form application offer initial solutions, they pose limitations regarding scalability and compatibility with emerging technologies. This potentially hinder long-term adaptability with other systems, necessitating consideration for future upgrades and expansions.

Okon, Eleberi, & Uka (2020) crafted a web-based digital repository tailored for scholarly materials and publications within a tertiary institution. The motive behind this endeavor stems from the necessity to address the persistent demand for an efficient, reliable, and easily accessible system for storing and retrieving scholarly materials. Such a system is envisioned to alleviate the burdens faced by academic staff and students, facilitating smoother day-to-day activities, particularly in swiftly obtaining relevant scholarly information. Leveraging open-source software and Object- Oriented Analysis and Design Methodology (OOADM), the system development process aims to deliver a sustainable solution. Validation tests indicate that the proposed system effectively addresses key challenges encountered in managing and sharing scholarly information within the institution.

Olusanya *et al.*, (2021) research focused on developing an institutional repository tailored for storing and retrieving information on graduate theses, thereby mitigating the risk of duplicating existing studies. Functional and non-functional requirements were identified, guiding the system's design using Unified Modeling Language (UML) tools. Web 2.0 technologies were employed for implementation. The study's outcomes demonstrate that the system furnishes an online institutional repository, enhancing the quality of graduate research by preventing redundancy.

Tsolakidis *et al.*, (2023) study detailed the adaptation of VIVO by the University of West Attica in Greece, wherein existing systems are integrated, and research-related data from various online sources is aggregated. The resultant solution functions as an academic research repository, ensuring consistency and enhancing the visibility of the university's research outputs. The novelty of this paper lies in its presentation of an ontology-based system for documenting institutional research, which can be customized to suit the specific needs of an institution. Additionally, it highlights the significance of web interface and information visualization in maximizing the utility of the repository.

Pushpendra (2024) outlined the creation and deployment of a Virtual Library System (VLS), an online platform intended to update and improve conventional library services. Employing cutting- edge technologies, the system delivers a smooth and engaging user experience by integrating functionalities like online cataloging, management of digital resources, user verification, and collaborative utilities.

System Architecture

Figure 1 below outlines the system architecture for the online student final year project repository system.

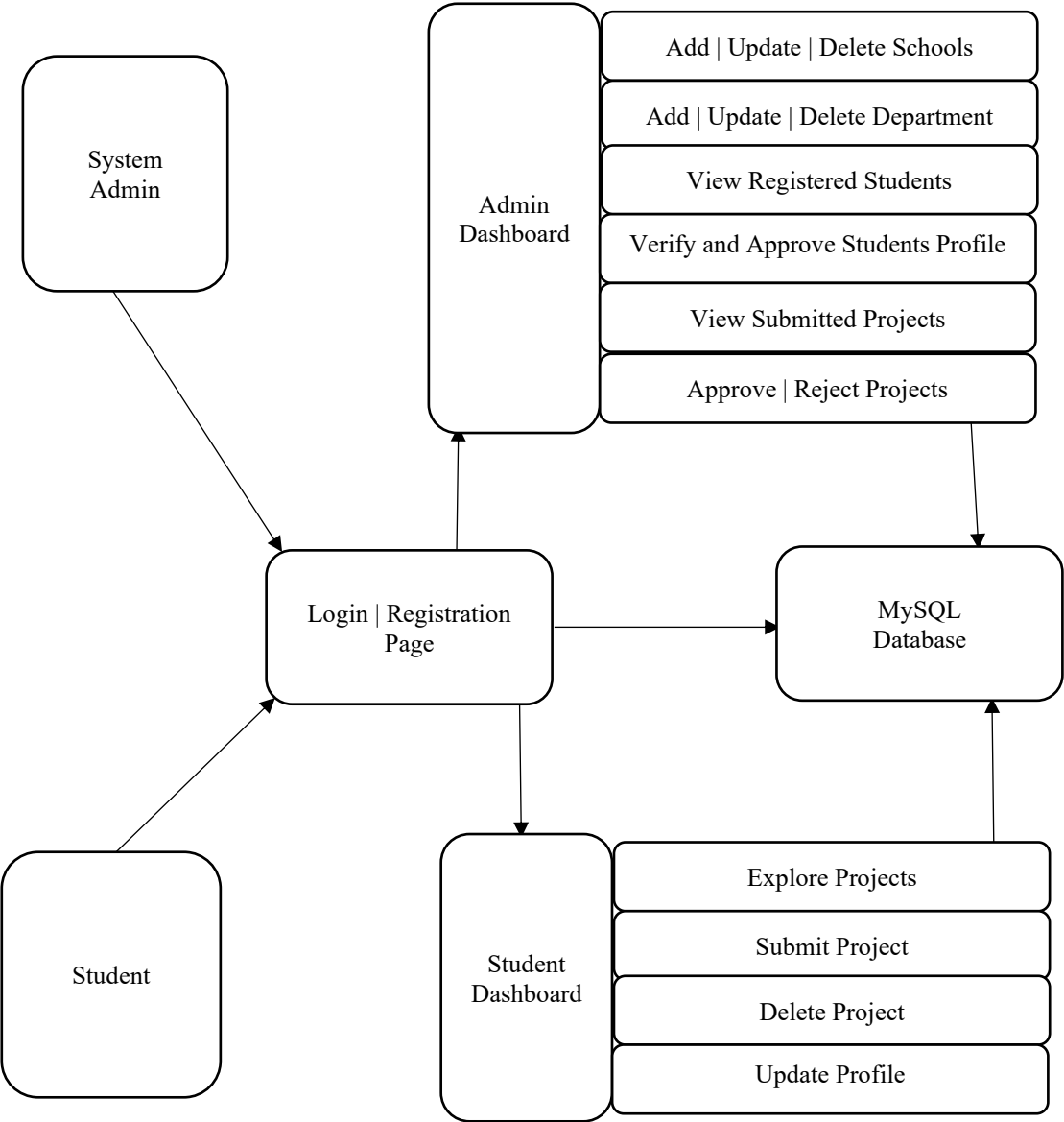


Fig 1. The System Block Diagram

The architecture includes several key components and their interactions, offering a detailed understanding of the system's functionality.

At the heart of the system are the System Administrators, responsible for managing various aspects, including schools, departments, projects, and user accounts. They access the platform through the Login/Register Page and navigate to the Admin Dashboard, where they can add,

update, or delete information about schools and departments, view submitted projects, approve or delete projects, and manage user accounts.

Student Users, on the other hand, focus on submitting and managing their projects. They also enter the system through the Login/Register Page and utilize the User Dashboard for their activities. This dashboard allows them to submit projects for approval, search and view existing projects, and edit or delete their submitted projects.

The Login/Register Page serves as the entry point for both System Administrators and Student Users, providing essential authentication and authorization services. Upon successful login, users are redirected to their respective dashboards. The Admin Dashboard for administrators and the User Dashboard for students.

The Admin Dashboard is equipped with functionalities that enable System Administrators to efficiently manage the system. They can add, update, or delete information about schools and departments, view all projects submitted by students, approve or delete projects, and manage user accounts by adding new users, updating user information, and deleting users when necessary.

The Student Dashboard offers Student Users the tools to manage their projects. They can submit new projects for review and approval, search for and view existing projects, and make edits or deletions to their submitted projects.

Central to this system is the MySQL Database, which serves as the repository for all data, including information about schools, departments, projects, users, and their interactions. The database interfaces with both the Admin Dashboard and the User Dashboard, ensuring that all actions performed by users are accurately stored and updated.

The interaction flow within this architecture begins with System Administrators logging in through the Login/Register Page and accessing the Admin Dashboard to manage system elements. Similarly, Student Users log in and access the User Dashboard to handle their project submissions. Both dashboards interact with the MySQL Database to perform data storage and updates, maintaining consistency and integrity across the system.

Requirement Analysis

Users should be able to search for student final year project based on keywords, authors, publication dates, journals, and other relevant metadata. Advanced search filters should be provided to facilitate precise searching and filtering of results. Users should have the ability to easily submit their student final year project to the repository. The submission process should include fields for title, authors, abstract, keywords, and affiliations. Support for uploading file formats (PDF) should be provided.

Administrators should have the ability to manage the content of the repository, including adding, editing, and deleting student final year project. Administrators should be able to manage user accounts, including creating, editing, and deactivating accounts. User roles and permissions should be configurable to control access to certain features or administrative functionalities.

System Design

Unified Modeling Language (UML) diagrams, such as use case diagrams, were utilized to illustrate the system's functionality from the perspective of the users. This means that it detailed individual system features and available functionalities as seen by the users. Figure 2a and 2b presents the use-case diagram of the proposed system.

System Administrator

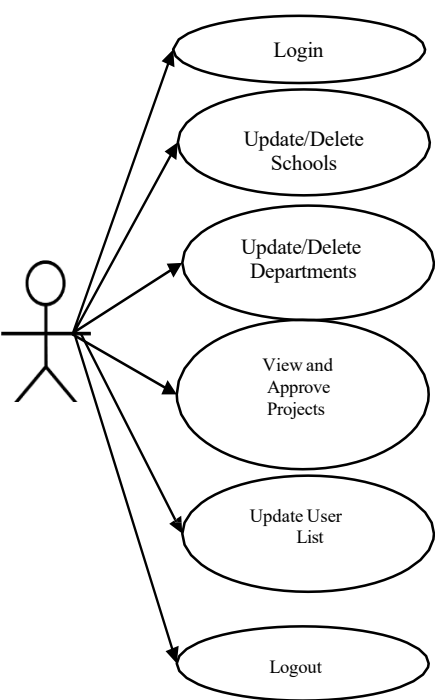


Fig 2a

Student

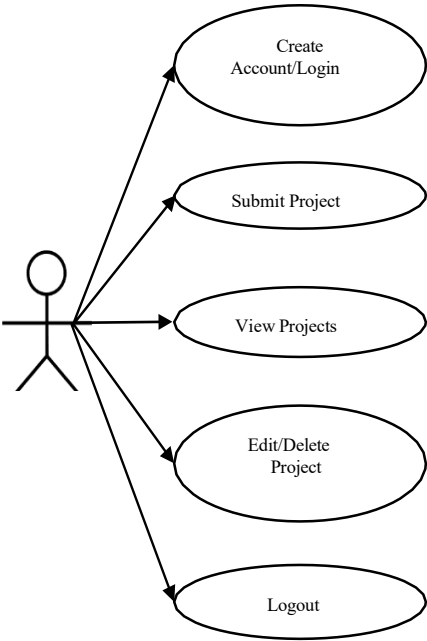


Fig 2b

Fig 2 Use Case Diagram

On the left side (Fig 2a), the use case diagram depicts a system administrator with six actions. The system administrator can log in, add, update and delete schools, add, update and delete departments, view submitted project works, approve and delete project works, add, update and delete user, and log out. On the right side of (Fig 2b), the diagram shows a student user with five actions. The student user can register or log in, submit project for approval, search for or view projects, edit/delete submitted project and log out.

Home Page

Figure 3 below shows the home page of the repository system that provides easy navigation for user to access the contents of the system.



Fig 3: Home Page

From the home page, user can navigate through the project section and filter by schools and departments as well login or register an account. The admin can access the login option the home page.

Admin Login Page

Figure 4 below shows the admin login page.



Fig 4: Admin Login Page

The Admin Login Page is a secure portal where administrators enter their credentials (username and password) to gain access to the administrative functions of the online academic repository system, ensuring only authorized personnel can manage the system.

Student Registration Page

The Student Registration Page as shown in figure 5 below allows new students to create an account by providing personal information, selecting a username and password, and agreeing to terms and conditions, enabling them to access and use the academic repository.

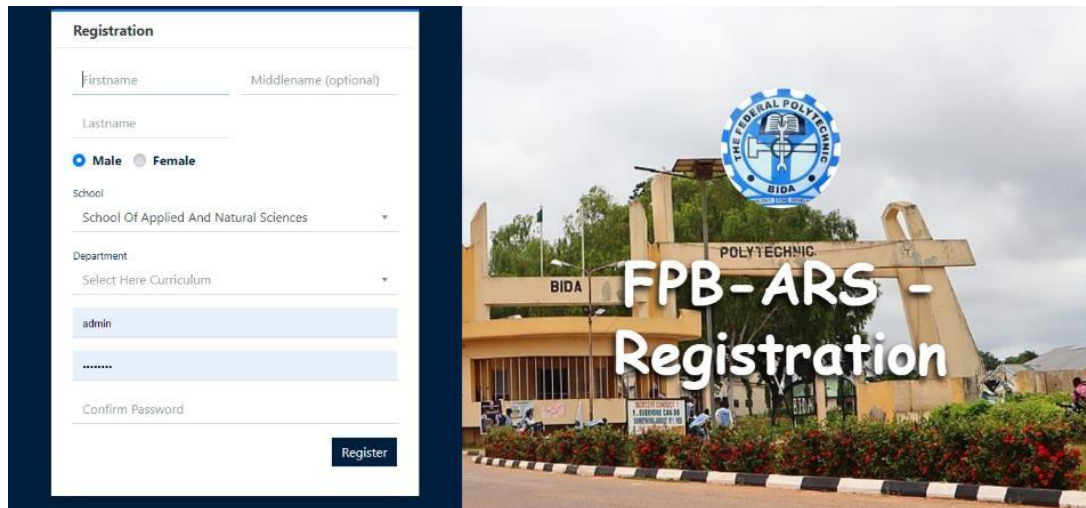
The figure consists of two parts. On the left is a screenshot of a web registration form titled "Registration". The form includes input fields for "Firstname", "Middlename (optional)", and "Lastname". Below these are radio buttons for "Male" (selected) and "Female". There is a "School" dropdown menu with "School Of Applied And Natural Sciences" selected, and a "Department" dropdown menu with "Select Here Curriculum" selected. At the bottom of the form are fields for "admin" (username) and "*****" (password), followed by a "Confirm Password" field and a "Register" button. On the right is a photograph of the entrance to the Federal Polytechnic Bida, featuring a large gate with the institution's logo and the text "FPB-ARS - Registration" overlaid in large white letters.

Fig 5: Student Registration Page

The system verifies whether the details being entered already exist in the database. If a match is found, the user will be notified of an existing account. Otherwise, the registration process will proceed, pending administrator approval before the user can log in.

Student Login Page

The Student Login Page as shown in figure 6 below is a secure access point where students enter their username and password to log in and access the resources and functionalities available to them within the academic repository system.

The figure consists of two parts. On the left is a screenshot of a web login form titled "Login". It has two input fields: the first contains the text "admin" and the second contains "*****". A "Login" button is located at the bottom right of the form. On the right is a photograph of the entrance to the Federal Polytechnic Bida, featuring a large gate with the institution's logo and the text "FPB-ARS - Login" overlaid in large white letters.

Fig 6: Student Login Page

The system cross-references the submitted details with the database. If both the username and password are correct, a session is created for the student, and the dashboard opens up for project submission.

Project Submission Page

The Project Submission Page shown in figure 7 below is where students upload their project files and provide necessary details for submission.

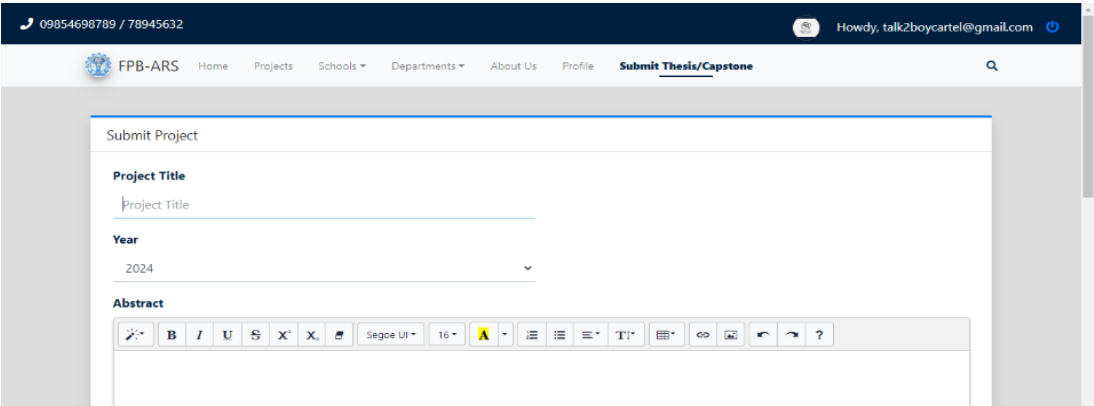


Fig 7: Project Submission Page

This page typically includes fields for entering the project title, description, and any relevant tags, as well as options to upload documents or other media files. The page ensures that students can submit their projects efficiently and that all required information is provided for review and assessment.

Admin Dashboard

The Admin Dashboard Page as shown in figure 8 below serves as the main interface for administrators after logging in, offering streamlined navigation to various essential sections.



Fig 8: Admin Dashboard Page

Administrators can access the school list to manage and update information about the schools within the system. The department list section allows for the addition and modification of departmental details under each school. The student list provides a comprehensive view of registered students, enabling administrators to manage student profiles, approve or deny registrations, and track their progress. Additionally, the project submitted list allows administrators to review, sort, and filter student project submissions, providing feedback or approvals as needed.

Project Explore Page

The All-Project Lists page displayed in figure 9 below is a comprehensive list of all submitted and approved projects within the academic repository system.

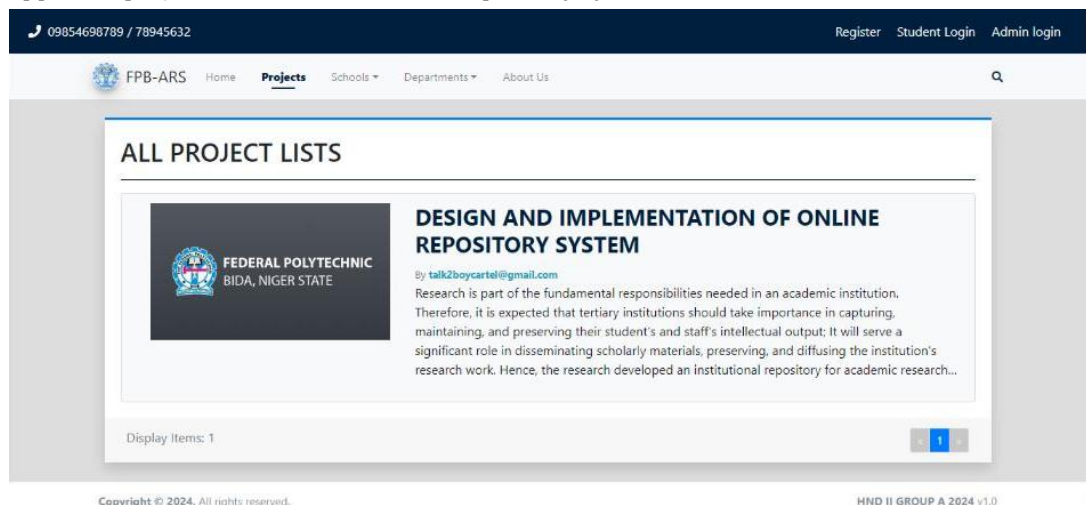


Fig 9: All Project Lists

This page typically includes project titles, authors, submission dates, and brief descriptions, along with filtering and sorting options to help users find specific projects or view them by category. This page allows both students and administrators to browse and access the repository's collection of projects efficiently.

Discussion

For the implementation, HTML, CSS, and JavaScript are employed for the frontend design to create a responsive and interactive user interface. PHP, alongside AdminLTE3, was used for the backend to handle server-side logic and user management functionalities. AdminLTE3 provides a modern and flexible framework for building the administrative interface, ensuring that system administrators can efficiently manage the repository. MySQL was used for the database to store and manage all the academic papers, user information, and other relevant data securely and

efficiently. The combination of these technologies ensures a robust, scalable, and user-friendly system, capable of handling complex queries and providing reliable data management.

CONTRIBUTION TO KNOWLEDGE

This research contributes to the academic community by developing a comprehensive online repository system that digitizes and centralizes final year projects, specifically designed for the Federal Polytechnic Bida. The repository enhances accessibility and usability of academic resources, addressing the limitations of the current manual system. By integrating modern technologies such as HTML, CSS, JavaScript, PHP, and MySQL, the system provides a scalable and efficient solution for managing academic works, facilitating easier access for students and administrators, and preserving valuable research outputs for future reference.

CONCLUSION

The design and implementation of an online academic repository system for final year student projects at the Federal Polytechnic Bida represent a significant advancement in the management and dissemination of academic research. This system not only addresses the inefficiencies and limitations of the previous manual processes but also provides a scalable, secure, and user-friendly platform for students, faculty, and researchers. By leveraging modern web technologies, the repository enhances accessibility, preservation, and the overall management of final year projects, thereby contributing to the broader academic and research communities.

RECOMMENDATIONS

While this research has successfully developed an online repository system, several areas were not covered and could benefit from further exploration and development:

Future work could explore integrating the repository with other academic systems, such as learning management systems (LMS) and institutional research databases, to create a more comprehensive academic ecosystem. Implementing advanced search functionalities using artificial intelligence (AI) and machine learning algorithms could further enhance the retrieval process, providing users with more relevant and precise search results. Developing a mobile application version of the repository would increase accessibility for users who prefer accessing academic materials on mobile devices, thereby expanding the system's reach and usability.

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